



TransLink Transit Intelligence Pipeline

INFO 4190- PROJECT INTEGRATION 1 | PROJECT PROPOSAL

Group 9

Inderpreet Kaur 100383066

Drishti Kapoor 100435109

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1. Project Title

TransLink Transit Intelligence Pipeline: System Design and Architecture for a Data Engineering and Machine Learning Solution for Public Transit Schedule Analysis and Commuter Departure Insights in Metro Vancouver

2. Team Composition and Roles

Full Name	Student ID	Role & Responsibilities
Inderpreet Kaur	100383066	Technical Design Lead, system architecture, data pipeline design, database schema, ML model design, API design, technical documentation
Drishti Kapoor	100435109	Design & Documentation Lead, UI/UX wireframes, mockups, requirements specification, literature review, progress reports, final report writing

3. Introduction & Problem Definition

Metro Vancouver's public transit network serves hundreds of thousands of daily commuters through buses, SkyTrain, SeaBus, and RapidBus services operated by TransLink. Planning a transit journey today is straightforward, apps like Google Maps and TransLink's Trip Planner tell you when the next bus arrives. However, none of these tools answer a different but equally important question: *when should you travel?*

A commuter heading from Surrey to Vancouver on a Sunday afternoon has no easy way to know whether noon is the busiest service window on their route, or whether leaving two hours earlier would result in a less frequent service journey. This kind of insight requires analysing hundreds of thousands of scheduled trips across the entire network, something no public-facing tool currently offers.

TransLink publishes its complete transit schedule as open data every week in the General Transit Feed Specification (GTFS) format. This dataset contains every route, stop, trip, and scheduled arrival across the Metro Vancouver network. Despite being publicly available, this raw data is entirely inaccessible to anyone without programming expertise, it arrives as multiple large, structured text files requiring significant technical knowledge to parse, join, and interpret meaningfully.

This project addresses that gap by producing a complete system design for an end-to-end data pipeline that would automatically download and process TransLink's weekly GTFS data, apply machine learning to identify service behaviour patterns, and present insights through a commuter-friendly interactive web dashboard. The design produced in this course will serve as a complete, implementation-ready blueprint to be built in the following semester.

The system is designed to answer questions real commuters actually ask: Which departure window on my route has the highest scheduled service frequency? Is Sunday afternoon

consistently busier in terms of scheduled trips than Sunday morning? When is the best time to travel? These answers would be presented in plain language, no technical knowledge required from the user.

Target users include Metro Vancouver daily commuters seeking smarter departure planning, transit researchers and urban planners studying network-wide service patterns, and open data enthusiasts interested in civic technology applications.

4. Literature Review

Several existing studies and transit platforms are relevant to this project. While each contributes valuable ideas related to public transit analytics, none directly addresses commuter-focused schedule intelligence for Metro Vancouver transit users.

[1] TransLink GTFS Open Data and Trip Planner

TransLink publishes its full weekly schedule as a free GTFS Static ZIP file containing 63,250 trips across 242 routes, and its Trip Planner shows real-time next departures for any stop [1]. But these two things don't connect for a regular commuter. The raw data requires programming skills to read, and the Trip Planner only shows the next few buses, it never tells you whether your chosen time is a busy or quiet window on your route. Our project bridges this gap by processing the open GTFS data and presenting what it reveals about schedule patterns in plain language.

[2] Antrim and Barbeau (2013) - *The Many Uses of GTFS Data*

Aaron Antrim and Sean Barbeau demonstrated that GTFS data can support applications far beyond traditional trip planning, including accessibility analysis, service mapping, multimodal planning, and transit visualization systems. Their work highlighted how open transit datasets create opportunities for innovative transportation research and public-facing applications.

The paper strongly supports the feasibility of using GTFS data for large-scale transit analysis. However, the applications discussed were mainly aimed at developers, planners, and researchers rather than everyday commuters. Our project builds upon this foundation by designing a complete end-to-end pipeline that converts GTFS schedule data into understandable commuter guidance presented through an interactive dashboard.

[3] Roth et al. (2011) - *Structure of Urban Movements: Polycentric Activity and Entangled Hierarchical Flows*

Camille Roth, Sung Moon Kang, Michael Batty, and Marc Barthélemy analysed London Underground Oyster card data to study how passenger movement patterns vary across an urban transit network. Their research found that transit scheduled activity is highly uneven, with a small number of routes and travel periods carrying a disproportionately large share of passenger movement.

This research directly supports the central assumption behind our project: transit activity varies significantly depending on route and time of day. Unlike the London study, however, TransLink does not publicly release passenger ridership datasets. Our system therefore uses scheduled

service frequency as a proxy indicator for service intensity, which is a common and academically accepted approach in transportation research.

[4] Transit App

Transit is widely used across North America and integrates GTFS and real-time transit feeds to provide live vehicle tracking, route planning, and departure predictions. Similar platforms such as Google Maps also focus primarily on navigation efficiency and real-time arrival information.

Although these systems are highly effective for trip navigation, they do not analyse historical or scheduled service patterns to help users determine the best time to travel. None provide plain-language insights such as identifying quieter travel windows, comparing scheduled service frequency trends across the day, or explaining schedule frequency behaviour at the network level.

a. Gap Addressed by This Project

Existing transit systems focus on answering the question: *“When is the next bus arriving?”*

This project instead addresses the question: *“When should I travel?”*

By processing publicly available GTFS schedule data through a structured data engineering and machine learning pipeline, the proposed system will provide commuters with understandable insights about service intensity, peak travel periods, and quieter departure windows across Metro Vancouver’s transit network. This type of commuter-oriented schedule intelligence is not currently available in existing public transit tools.

5. Proposed Methodology and Technologies

a. Design Approach

This project follows a structured waterfall design process appropriate for a design-only course. The process moves sequentially through requirements gathering, research, conceptual design, detailed system design, and UI/UX prototyping, with each phase producing documented design artefacts that feed directly into the next. This approach ensures that by the end of the course, a complete, coherent, and implementation-ready design package exists covering every layer of the proposed system.

b. Design Activities

Requirements and research (Weeks 1–6): Gather and document functional and non-functional requirements, research existing solutions, analyse GTFS data structure, and define system scope and constraints. Conceptual and system design (Weeks 7–9): Produce system architecture diagram, data flow diagrams, database schema, API endpoint specification, and ML model design. UI/UX design (Week 10): Produce wireframes and high-fidelity mockups for all five dashboard pages. Review and finalisation (Weeks 11–13): Peer review, design refinement, and final report and presentation.

c. Proposed Technologies for Implementation

The following technologies are recommended for the implementation phase next semester. Each is selected based on suitability for the problem, industry adoption, and alignment with the team's existing skills:

Technology	Layer	Justification
Python 3.x	Data pipeline and ML	Industry standard for data engineering; extensive library ecosystem for ETL and machine learning; free and open source
Pandas	Data processing	Efficient handling of large CSV-format GTFS files; supports memory-efficient column-selective loading for files with millions of rows
Scikit-Learn (K-Means)	Machine learning	Well-documented unsupervised clustering; requires no labelled training data; reproducible and academically valid for transit pattern analysis [2]
PostgreSQL	Database	Production-grade relational database appropriate for structured GTFS data, repeated imports, relational queries, API access, filtering, and dashboard use. If location-based stop search features are implemented, PostgreSQL + PostGIS will be used to support geospatial queries.
Flask	Backend API	Lightweight Python web framework; cleanly separates data logic from the presentation layer; low overhead for a two-person team
Chart.js	Frontend dashboard	JavaScript charting library used with Flask templates to build the interactive web dashboard
GitHub	Version control	Enables team collaboration, tracks design document versions, and supports public portfolio visibility

6. Expected Results and Practical Implications

By the end of Week 13, this project will produce the following design deliverables, all ready for implementation in the following semester:

- A complete requirements specification documenting all functional requirements, non-functional requirements, constraints, and assumptions for the system.
- A system architecture diagram showing how the ETL pipeline, PostgreSQL database, Flask API, K-Means ML model, and chart.js connect and communicate.
- A data flow diagram tracing the journey of TransLink GTFS data from download through processing, storage, analysis, and presentation.
- A database schema defining all tables, columns, data types, primary and foreign keys, and relationships derived from the GTFS Static file structure.

- A ML model design document specifying the K-Means clustering approach, input features, expected cluster count, evaluation method, and how cluster results feed into the dashboard.
- A Flask API specification defining all planned endpoints, their inputs, outputs, and the data they serve to the frontend.
- Wireframes and high-fidelity UI/UX mockups for all five planned dashboard pages: Network Overview, Route Analyser, Stop Explorer, Trip Planner Lite, and Network Patterns.
- A final report documenting all design decisions, rationale, known limitations, and a recommended implementation roadmap for the development phase next semester.

a. Features

Core Features

The following features are required and will be the primary focus of the implementation semester:

Core Feature	Description
GTFS Import Pipeline	Import core GTFS files including stops, routes, trips, stop_times, and calendar into PostgreSQL
Route Search	Allow users to search for a route by number or name
Stop Explorer	Allow users to search for a stop and view scheduled service patterns
Route Analyzer	Display schedule frequency by day, time window, and direction
Dashboard Filters	Allow filtering by route, stop, day of week, time range, transit mode, and direction
Service Intensity Charts	Visualize scheduled service frequency patterns
Plain-Language Summary	Provide a user-friendly explanation of what the chart means
Data Refresh Status	Show when the GTFS feed was last processed

Optional / Stretch Features (Implemented if time permits)

Stretch Feature	Description
Nearby Stop Search	If using PostGIS, users can search for stops near a selected location
Compare Two Routes	Users can compare two possible routes or two departure windows

Data Refresh Status Page	Shows when GTFS data was last imported and whether the pipeline ran successfully
Admin / Pipeline View	Shows imported files, row counts, ETL status, errors, and processing logs
Export Results	Allows users or researchers to export route analysis results as CSV or PDF
Saved Routes or Common Searches	Allows users to save frequently used routes or stops

Practical implication: Once built from this design, a Metro Vancouver commuter could open the Route Analyser, select their route and day, and immediately see whether their planned departure falls in a peak or off-peak service window, with a plain-language suggestion to travel at a quieter time if one exists. This insight is currently unavailable through any existing public TransLink tool.

Design limitation acknowledged: The proposed system infers service intensity from schedule frequency as a proxy since TransLink does not publish ridership data publicly. Higher scheduled frequency indicates TransLink's own planning data supports higher demand in that window. This methodology is consistent with standard academic transit research practice and will be documented transparently in all design artefacts [2].

7. Project Timeline

Week	Tasks and Deliverables
Week 1	Team formation, project topic confirmation, initial exploration of TransLink GTFS dataset
Week 2	Proposal drafting, literature review research, proposal submission
Week 3	Github Repo Setup
Week 4	Detailed analysis of GTFS file structure and identification of system requirements
Week 5	Initial system architecture design and database schema planning
Week 6	Peer Review 1, feedback incorporation, refinement of architecture and requirements
Week 7	Design of ETL pipeline workflow and creation of data flow diagrams
Week 8	Completion of database schema and Flask API endpoint specifications
Week 9	Machine learning model design, clustering methodology documentation, evaluation planning
Week 10	UI/UX wireframes and high-fidelity dashboard mockups for all planned pages
Week 11	Integration of all design artefacts into the final system documentation
Week 12	Peer Review 2, final revisions, risk review, implementation roadmap refinement

Week 13	Final report completion, presentation preparation, and final project submission
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8. Risk Assessment and Mitigation Plan

Risk	Likelihood	Mitigation
GTFS data structure is more complex than anticipated, making database schema design difficult	Medium	Conduct thorough GTFS data exploration in Week 1 before any design decisions are made.
System design scope becomes too large to complete all five dashboard pages within 13 weeks	Medium	Trip Planner Lite is the lowest priority page. If time is limited, the core three pages, Network Overview, Route Analyser, and Network Patterns, constitute a complete and presentable design on their own.
Team coordination challenges result in inconsistent design artefacts across sections	Medium	Both team members agree on design standards and naming conventions in Week 2. All artefacts stored in a shared GitHub repository so both members can review each other's work before submission.
Peer review feedback in Week 6 or Week 12 requires significant redesign	Low	Design decisions are documented with clear rationale from Week 4 onward, making it straightforward to explain and defend choices. Minor revisions are expected and built into Weeks 11–12 as a buffer.

9. Success Metrics and Evaluation Plan

a. Design-Phase Metrics

Metric	Target
Requirements specification covers all functional and non-functional requirements	Minimum 10 functional requirements and 5 non-functional requirements documented and reviewed by both team members
System architecture diagram accurately represents all proposed system components and their connections	Reviewed and approved by both team members; no missing components identified during Peer Review 1
Database schema correctly maps all relevant GTFS Static files to relational tables	All core GTFS files, stops, routes, trips, stop_times, calendar, represented with correct keys and relationships

UI/UX mockups cover all five planned dashboard pages	All five pages completed with annotated wireframes and at least one high-fidelity mockup per page
Design is assessed as feasible during Peer Review 2	No major feasibility concerns raised that cannot be addressed within the design's stated constraints
Final report clearly documents all design decisions and implementation roadmap	Report reviewed by both team members and submitted on time by August 4, 2026

b. Implementation Phase Success Metrics

Metric	Target
GTFS import pipeline runs successfully	All core GTFS files (stops, routes, trips, stop_times, calendar) import into PostgreSQL without errors
Route Search and Stop Explorer features work correctly	Users can search by route number/name and stop name and receive accurate schedule results
Service Intensity Charts render correctly	Charts display scheduled frequency data accurately for the selected route, day, and direction
Dashboard load time	Core pages load within 3 seconds on a standard broadband connection
Plain-language summaries are generated	Every Route Analyser result includes a plain-language description of the service pattern
Data Refresh Status displays correctly	Dashboard shows the date the GTFS feed was last processed and whether the pipeline ran successfully

10. References

- [1] TransLink, "GTFS Static Data," South Coast British Columbia Transportation Authority, 2026. [Online]. Available: <https://www.translink.ca/about-us/doing-business-with-translink/app-developer-resources/gtfs/gtfs-data>
- [2] A. Antrim and S. J. Barbeau, "The many uses of GTFS data - opening the door to transit and multimodal applications," Location-Aware Information Systems Laboratory, University of South Florida, 2013. [Online]. Available: <https://www.researchgate.net/publication/268097444>
- [3] C. Roth, S. M. Kang, M. Batty, and M. Barthélemy, "Structure of urban movements: Polycentric activity and entangled hierarchical flows," *PLOS ONE*, vol. 6, no. 1, e15923, Jan. 2011. doi: 10.1371/journal.pone.0015923
- [4] Transit Inc., "Transit - the best app for buses and trains," 2026. [Online]. Available: <https://transitapp.com>